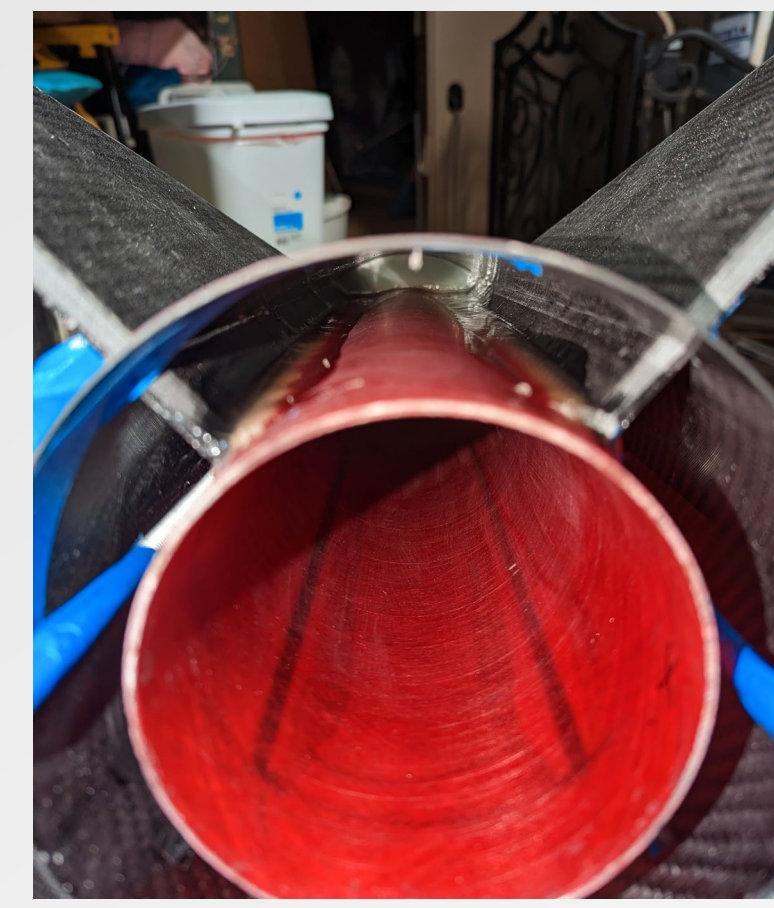


STRUCTURES

Karkinos is a 6" diameter, 12.9 ft tall SRAD rocket. Its primary structures are made from fiberglass and carbon fiber. The fin can is entirely custom including a hand rolled carbon fiber airframe and four fins. Each fin has a 3/16" G10 fiberglass core with 2 layers of carbon fiber on each side. The fin can contains a 98mm motor mount, with a custom aluminum thrust plate and a commercial retainer. The remaining airframe tubes are hand rolled fiberglass tubes to allow for signal transparency. The tubes and fin can were topped with a layer of peel ply and laminating epoxy, followed by clearcoat for the fin can and primer and paint for the tubes.

The couplers, motor mount tube, and nosecone are commercial filament wound fiberglass. Karkinos also features multiple custom aluminum bulkheads for the electronics bay and nosecone. Tube connections are made through 1/4-20 bolts and 4-40 shear pins.



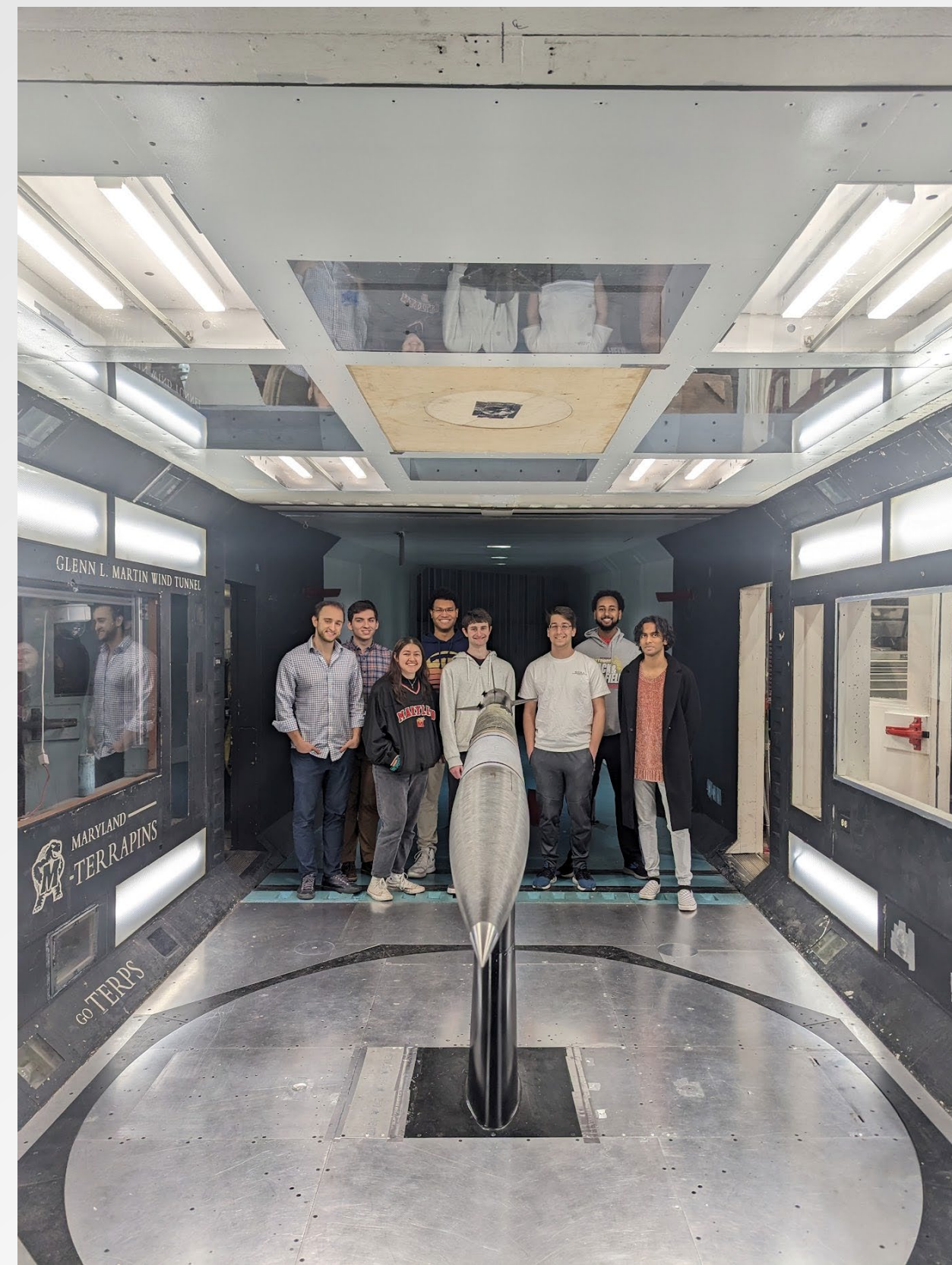
Above: Internal Fillets
Left: Fin Can



KARKINOS

UNIVERSITY OF MARYLAND:

TERRAPIN ROCKET TEAM



Team members with Karkinos in the Glenn L. Martin Wind Tunnel



Members of the Terrapin Rocket Team display Karkinos in Washington, DC



Team members at the pad for Karkinos' 4th flight



Karkinos on its four test flights, in order from left to right

TEST FLIGHTS

Karkinos flew four test flights, all taking place at MDRA in 2023.

- 01/07/23 - Loki M3464 to 8300 ft – First flight of new airframe and recovery systems
- 02/04/23 - M3000 to 8500 ft - Improved recovery systems, weather related issues, durability concerns of airbrake hinge
- 03/05/23 - EX N2500 to 10,330 ft – First flight with experimental motor, payload successfully deployed
- 04/02/23 - EX N2500 to 9600 ft – successful airbrake deployment, avionics nominal

These test launches helped the team become familiar with the rocket and launch procedures. The team gathered vast quantities of data that helped shape future control systems of the custom avionics and airbrake.

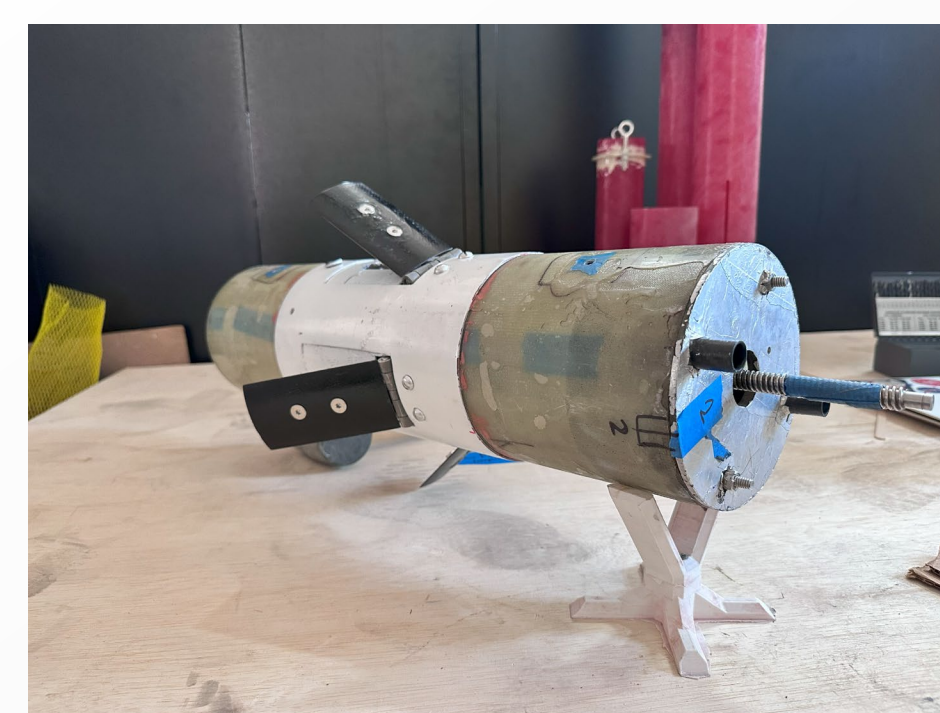


AIRBRAKE

The air brake module is designed to induce drag with deployable flaps to reach a desired apogee. The structure is built from hand-rolled fiberglass tubes, waterjet cut aluminum plates, and 3D printed parts. The air brake is designed to be modular and sits above the fin can in the full assembly of the rocket. An SRAD flight computer continuously predicts the trajectory of the rocket during ascent and deploys the flaps to precise angles using a stepper motor.

In its second year of development, the airbrake went through significant research and development. Primarily, in February 2023, the rocket was placed in the Glenn L. Martin Wind Tunnel at the University of Maryland to characterize the induced drag of the air brake flaps and to validate our physics models. This, along with data from nine flights of the computer, helped shape the control algorithms of the module. The wind tunnel testing was crucial to the development of the system. In addition, numerous upgrades were made to the air brake. Rebuilding and redesigning resulted in a 20% weight reduction and 3D printed steel flap hinges for strength. For software, the SRAD avionics package upgraded the communication protocol, control loop frequency, and data fusion algorithm.

The air brake was fully integrated and tested in Karkinos on 4/2/23. It shaved off nearly 800 ft of the predicted apogee and came within 73 ft of the desired apogee!



The airbrake module with its flaps deployed



Karkinos with the airbrake flaps actuated during tests at the Glenn L. Martin Wind Tunnel at UMD

AVIONICS

For the flight computers in Karkinos, dual Altus Metrum EasyMinis were chosen as the primary and backup. The EasyMinis provided dual deployment capabilities, while a Featherweight GPS tracker provided live telemetry and GPS data. This allows the team to monitor the health of the rocket while on the pad and in flight. The Altus Metrum TeleGPS was also onboard as a backup GPS tracker.

In addition to commercial flight computers, student designed avionics were also flown. The purpose of custom avionics is to develop a high fidelity flight computer that can perform datalogging and telemetry to a ground station. The completely student designed board is currently capable of logging altitude, orientation, acceleration, and GPS position, which are stored in an SD card. It is also capable of transmitting information including altitude, speed, and position to a ground station via radio. This is accomplished by transmitting APRS packets over LoRa protocol to a ground receiver. Flight tests were successful, confirming the 2 mile range. Future improvements include transitioning from breakout boards to SMDs and upgrading the radio system to transmit more data over greater distances.

RECOVERY

Karkinos utilizes standard two break dual deploy for its recovery system. At apogee, a commercial 24" drogue is deployed and the rocket descends at a rate of 104 ft/s. At 1000 ft above the ground, the main parachute, a SkyAngle C3/XXL, its deployed to slow the rocket down to a safe speed of 30 ft/s. The parachutes are wrapped in nomex blankets and shock cord is made of tubular Kevlar.

PAYLOAD

The payload of Karkinos is housed within the nose cone. It is a 9 lb, 4U cubesat structure designed to release an inner vehicle. It contains a microcontroller with sensors including an altimeter and IMU to help it understand orientation. A cable keeps the vehicle connected to the payload, but at 1200 ft during descent, servos on the payload actuate springs that release the vehicle. The test vehicle is a cylinder connected to a parachute. The goal is to steer the parachute by manipulating the shroud lines with servos. The vehicle contains another microcontroller with its own onboard GPS, accelerometer, and altimeter.



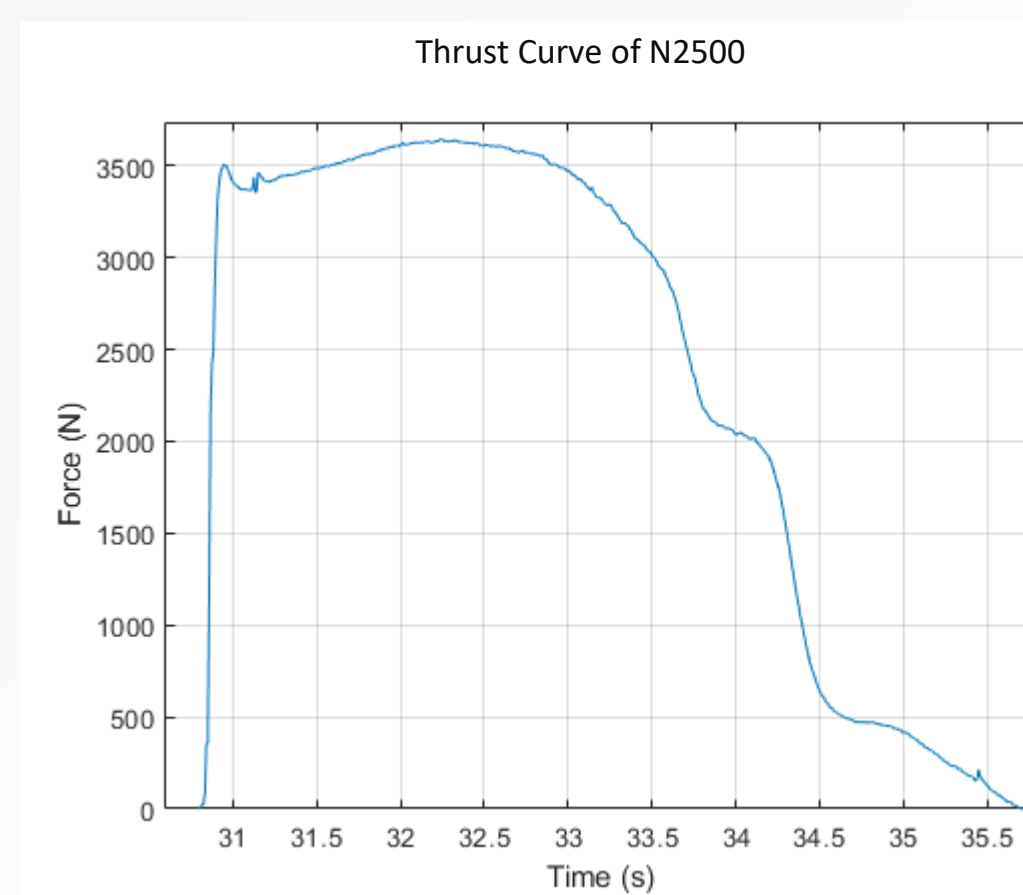
Above: The release vehicle integrated into the payload and nose cone coupler.
Left: The release vehicle parachute after a successful deployment and landing.

PROPULSION

Karkinos will be flying on an experimental N2500 motor. It was designed to have a total impulse of about 11,500 Ns and a peak thrust of about 3,600 N. This 98mm 7 grain APCP motor was designed by the team under the guidance of experienced mentors. The casings and closures were COTS. It was test fired twice to gather data before flying twice.

Karkinos used a modified version of a common propellant formula named Terple Nebula. Careful planning, caution, and mentorship were crucial to ensuring the safety of the mixing and casting process. The team static fired 6 smaller test motors to characterize the propellant, and then proceeded to mix motors for the full scale static fire and flights.

The 98mm casing was made of 6061-T6 Aluminum from Fisher Research, and the liner was an XX phenolic liner from Franklin Fibre. The graphite nozzle and aluminum forward closure were also produced by Fisher.



Above: Thrust Curve of N2500
Left: One of two static fires of the experimental motor